

Multiple Choice (10 marks)

1. Keiko spent the day bird watching and counted 34 more birds in the morning than in the afternoon. If she counted a total of 76 birds, how many birds did she count in the afternoon?
 - (a) 84 birds
 - (b) 45 birds
 - (c) 21 birds
 - (d) 34 birds
 - (e) 42 birds
2. If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel?
 - (a) 90 miles
 - (b) 120 miles
 - (c) 150 miles
 - (d) 140 miles
 - (e) 26 miles
3. You and three friends go to a concert. The total cost for four tickets is \$112. Write and solve an equation to find the cost of one ticket t .
 - (a) $4 - t = 112$; -\$108
 - (b) $4t = 112$; \$28
 - (c) $t \text{ over } 4 = 112$; \$448
 - (d) $t + 4 = 112$; \$108
 - (e) $t \text{ over } 4 = 112$; \$28
4. what is the value of $\int_a^b \frac{dx}{\sqrt{(x-a)(b-x)}}$? Round the answer to the thousands decimal.
 - (a) 0.6931471
 - (b) 3.7320508
 - (c) 2.3025851
 - (d) 6.2831853
 - (e) 1.4142135
5. What is the order of the element 5 in U_{48} ?

- (a) 3
- (b) 4
- (c) 1
- (d) 5
- (e) 9

True / False (10 marks)

Answer True or False

6. True or False: The sum of the roots of the equation $x^3 - 2x + 2 = 0$, when calculated correctly, is 1.
7. True or False: In the number 462,017, the value of the digit 6 is exactly 10 times the value of the digit 6 in the number 186,425.
8. True or False: The sum of the roots of the equation $x^3 + 3x^2 - 4 = 0$ is 3.
9. True or False: Using the Birge-Vieta method, the equation $x^3 - 11x^2 + 32x - 22 = 0$ has no real roots.
10. True or False: The value of A for which the quartic $x^4 + 3x^3 + 11x^2 + 9x + A$ has roots k, l, m, and n with $kl = mn$ is 9.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Evaluate $\sqrt[3]{12} \times \sqrt[3]{20} \times \sqrt[3]{15} \times \sqrt[3]{60}$.
12. Given matrices A (2×7) and B (7×5), analyze the process of matrix multiplication to determine the dimensions of the resulting product matrix AB and explain your reasoning.

End of Paper